

PARIMALA SUMA SANJANA, CHENJERI

+1 (682) 374-4752 | sumasanjana.ch@gmail.com | [linkedin.com/in/sanjana-suma](https://www.linkedin.com/in/sanjana-suma)

EDUCATION

Master of Science (M.S.) – Computer & Information Science
Southern Arkansas University, Magnolia, AR | GPA: 3.85

Aug'24 - July'25

PROFESSIONAL EXPERIENCE

Systems Engineer – Infosys Ltd. (Client: SSE, UK)

Jun'21 – Jun'22

- Engineered **Python-driven SQL debugging framework** that improved query accuracy from 99.7% to 99.99% and cut error resolution time by 30%, ensuring higher data reliability in reporting pipelines.
- Designed and deployed **Java-based ETL components** using **object-oriented design principles**, which streamlined enterprise **data pipelines** and reduced duplication defects by 15%, enhancing maintainability across five major applications.
- Implemented a **data validation framework** that detected 95% of anomalies before production, preventing critical **financial reporting** issues and guaranteeing zero **data loss** across business units.
- Accelerated delivery cycles by 15% through adoption of **Agile practices** and **CI/CD pipelines**, enabling continuous integration, improved sprint velocity, and faster deployment of software modules.
- Collaborated with engineers to execute **AWS migration project**, ensuring zero downtime, **GDPR compliance**, and enhanced scalability, which resulted in recognition with the **Rising Star award** in Q4 2021.
- Resolved 95% of software bugs in five modernization upgrades within four months, achieving improved stability, reduced latency by 30%, and strengthened end-user system performance.
- Conducted **technical interviews** for 11 candidates, expanding the **engineering team** and contributing to increased project bandwidth, which directly supported timely modernization deliverables.

INTERNSHIPS

Tapsoft Tech Labs Pvt. Ltd. (Full Stack Developer)

Aug'20 - May'21

Healthcare Claims Data Processing & Analytics Pipeline:

- Built **Python** and **SQL-based data cleansing pipelines** that standardized diagnostic codes (**ICD-10, CPT**) and resolved missing values, improving dataset accuracy by 25% for **healthcare analytics**.
- Developed **RESTful APIs** with **Spring Boot** to automate course data integration, reducing manual entry errors by 15% and saving the training division 10+ hours weekly.
- Constructed a **digital badge certification platform** that distributed 12+ badges weekly, increasing workforce engagement by 40% and reducing onboarding complexity across PwC learning teams.
- Optimized **SQL queries** and migrated them to **efficient window functions and CTEs**, lowering CPU usage by 70% during peak reporting periods and stabilizing **analytics dashboards**.
- Proposed and documented a transition to **AWS S3-based architecture**, which was unanimously approved by leadership, enabling scalable storage, cost reduction, and improved analytics adoption.

Oil and Natural Gas:(Network Security Intern)

May'19 - Jun'19

- Ran penetration testing** on telecommunication systems, identifying and **rectifying 18 critical vulnerabilities**, thereby **mitigating potential work exploits and data breaches**, and upgrading **system resilience**.
- Bolstered network security infrastructure** by **deploying detection systems (IDS)** and **establishing comprehensive security policies**, averting **95+ attempted breaches monthly** and **safeguarding critical customer information**.

UNIVERSITY PROJECTS:

- Multiplayer Battleship – Python Game:** Prepared a **multiplayer** game using **Python, Tkinter, Pygame, and Streamlit**, incorporating **sophisticated algorithms** to ensure **valid ship placements** and prevent cheating, resulting in **zero reported exploits**.
- Multi-Cloud Security Infrastructure:** Implemented **fault-tolerant multi-cloud architecture** across **AWS, Azure, and GCP**, configuring **IAM, VPCs, and firewall rules**, bolstering **network security** by 60% and reducing **security incidents** by 35%.
- University PHP Academic Portal:** Conceptualized and delivered an intuitive **PHP-based academic portal** for **student records management**, enabling faculty to process **100+ updates daily** and contributing to a **20% reduction in data entry errors**.
- E-commerce Review Sentiment & Risk Data Pipeline:** Designed and operated an **end-to-end data pipeline** for e-commerce reviews, enabling the **product development team** to address the **three most common customer complaints** within one quarter.
- Arlington Sprouts – Full-Stack Web Application:** Built a web application featuring **real-time updates** using **React and PostgreSQL**, achieving a **15% query efficiency gain** verified through **Postman tests** within the first four weeks.

TECHNICAL SKILLS:

Programming Languages: Python, Java, C, JavaScript, Bash/Shell.

Frameworks/Tools: HTML/CSS, Microservices, Webpack, Babel, Postman, Git, Data Bricks, Airbyte, DBT.

Databases: PostgreSQL, MySQL, NoSQL, MongoDB.

Cloud & DevOps: AWS (Amazon Web Services), Azure, GCP (Google Cloud Platform), Docker, Kubernetes.

